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CS-499 Module 3 Milestone Two

Artifact Enhancement 1: Design and Engineering

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The artifact I selected for Milestone Two was the Angular administrative portion of my Travlr application. I selected this artifact because it already included authentication, routing, and CRUD functionality, which made it a strong candidate for software design and engineering enhancements. The original application functioned correctly, but much of the authentication logic was duplicated across multiple components, and the form validation and error handling were limited. I wanted to improve the maintainability, usability, and overall structure of the application while keeping the enhancement realistic for a student-level project.

The primary enhancement I implemented was the addition of Angular route guards to centralize route protection for administrative pages. Previously, components such as Trip Listing, Add Trip, and Edit Trip each contained their own authentication checks and redirect logic. I replaced this duplicated logic with a reusable authGuard that protects routes directly through Angular routing. This improvement created better separation of concerns because the routing layer now handles access control while the components focus only on application functionality and user interaction. This enhancement improved maintainability and made the application easier to scale and manage.

In addition to the route guard enhancement, I improved form validation and error handling in the Add Trip and Edit Trip features. I added stronger reactive form validation, including required field checks, minimum length validation, and numeric validation for pricing fields. I also added clearer validation messages within the user interface so users receive immediate feedback when entering invalid data. These changes improved both the user experience and the quality of the application design.

This enhancement aligns with several course outcomes. The project demonstrates the ability to apply software engineering practices to improve maintainability, modularity, and

usability within an existing application. The implementation of reusable route guards and centralized authentication logic reflects stronger software design principles and improved organization of application responsibilities. The enhanced validation and error handling also demonstrate the ability to develop more reliable and user-focused software solutions.

Through this enhancement process, I learned more about Angular routing, route guards, reactive forms, and centralized authentication handling. I gained experience improving an existing codebase instead of only creating new functionality. One of the biggest challenges I faced was restructuring the authentication flow without breaking existing application behavior. Because the original components handled their own authentication checks, I needed to carefully move that logic into routing while ensuring all protected pages still behaved correctly. Another challenge was implementing validation improvements consistently across multiple forms while maintaining compatibility with the existing application structure. Successfully completing these enhancements improved my understanding of software design and reinforced the importance of maintainable architecture and reusable components.